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Exercise 4a – Prototype with Familiar and Unfamiliar Navigation Elements

Title: Student Learning App – Mobile Prototype

AIM

The aim of this exercise is to design and develop a mobile learning application prototype using Proto.io that combines both familiar and innovative navigation elements. The usability of these elements is evaluated among different user groups to understand ease of use and interaction efficiency.

OBJECTIVE

- To design a simple and intuitive student learning application.
- To incorporate familiar navigation patterns for easy usability.
- To introduce minimal modern UI elements to enhance engagement.
- To evaluate how different users respond to these navigation styles.

PROCEDURE

Step 1: Planning the Prototype

Before designing in Proto.io, the structure of the application was planned.

Identified Navigation Elements

Familiar Navigation Elements:

- Home screen with main menu buttons

- Subject category grid layout
- Search bar
- Back button navigation
- Profile information screen

Unfamiliar / Modern Navigation Elements:

- Card-style subject layout
- Minimalist centered layout design
- Gradient-style buttons
- Clean spacing and simplified navigation flow

Step 2: Creating the Project in Proto.io

1. Logged into Proto.io.
2. Clicked on **Create New Project**.
3. Selected mobile device (iPhone model).
4. Named the project **Student Learning App**.
5. Created blank screens for design.

SCREENS INCLUDED

The prototype consists of three main screens:

1. Home Screen
2. Courses Screen
3. Profile Screen

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1. Home Screen

Features:

- App title: “Student Learning App”
- Image banner at the top
- Three main buttons:
    - Courses
    - Profile
    - Settings

Navigation Type:

- Vertical button layout
- Large clickable buttons
- Clear text labels

Familiar Elements:

- Menu-style button layout
- Clear call-to-action buttons
- Standard home screen structure

Purpose:

To provide a simple starting point where students can easily access different sections.

2. Courses Screen

Features:

- Title: “Explore Courses”
- Search bar at the top
- Subject cards:
    - Tamil
    - English
    - Maths
    - Science
    - Social
- Home back navigation button

Familiar Navigation:

- Search bar for quick access
- Grid-based subject selection
- Back/Home navigation button
- Clearly labeled subject names

Unfamiliar / Modern Elements:

- Card-style subject buttons
- Minimalist UI with clean spacing
- Gradient color theme for subject boxes

Purpose:

To allow students to quickly choose their subject of interest in a visually organized manner.

3. Profile Screen

Features:

- Profile picture
- Personal Details button
- Professional Details button
- Settings button

Familiar Navigation:

- Profile picture at top
- Section-based details
- Clear button navigation

Unfamiliar / Modern Elements:

- Large rounded profile image
- Color-coded buttons (Green & Purple theme)
- Simplified layout with focused sections

Purpose:

To give students access to their personal and academic information in an organized way.

FAMILIAR NAVIGATION ELEMENTS USED

These elements improve ease of use:

- Home screen main menu
- Search bar
- Back/Home navigation
- Profile section
- Clearly labeled buttons
- Grid subject layout

These patterns are commonly used in educational and mobile applications, making them intuitive for most users.

UNFAMILIAR / INNOVATIVE ELEMENTS USED

To enhance engagement and modern design:

- Card-style subject layout
- Gradient buttons
- Minimalistic centered UI
- Large rounded profile image
- Simplified navigation flow

These elements aim to make the app visually appealing while maintaining usability.

USER GROUPS FOR USABILITY TESTING

Users were divided into three groups:

1. **School Students (13–17 years)**
    - Basic technical knowledge
    - Prefer visual and simple navigation
2. **College Students (18–22 years)**
    - Tech-savvy
    - Comfortable with modern UI
3. **Teachers (25+ years)**
    - Moderate technical experience
    - Prefer structured and clear navigation

USABILITY TESTING PROCESS

Tasks Given:

- Open the app and navigate to Courses
- Search for a subject
- Return to Home screen
- Open Profile page
- Access Personal Details

Testing Method:

- Shared Proto.io preview link

- Observed task completion time
- Collected feedback using a short questionnaire

OBSERVATION

Familiar Elements:

- All users easily identified main buttons.
- Search bar was quickly understood.
- Home navigation was simple and clear.

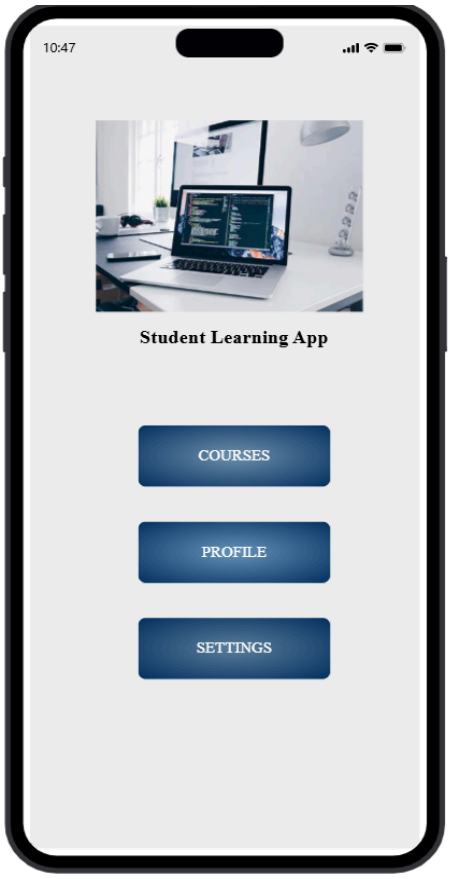
Novel Elements:

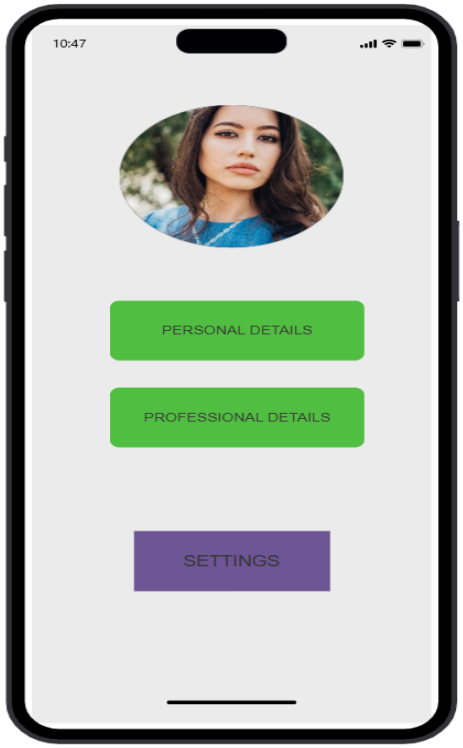
- Younger users liked the card-style layout.
- Some teachers preferred more traditional menu style.
- Gradient buttons were visually attractive but not confusing.

ANALYSIS

- Familiar navigation ensured smooth usability.
- Card-based layout improved visual clarity.

10:47
EXPLORE COURSES:
Search
TAMIL
ENGLISH
MATHS
SCIENCE
SOCIAL
HOME


10:47
Student Learning App
COURSES
PROFILE
SETTINGS


10:47
PERSONAL DETAILS
PROFESSIONAL DETAILS
SETTINGS